NOVA SCHOOL OF BUSINESS & ECONOMICS

Economics of education

2nd class

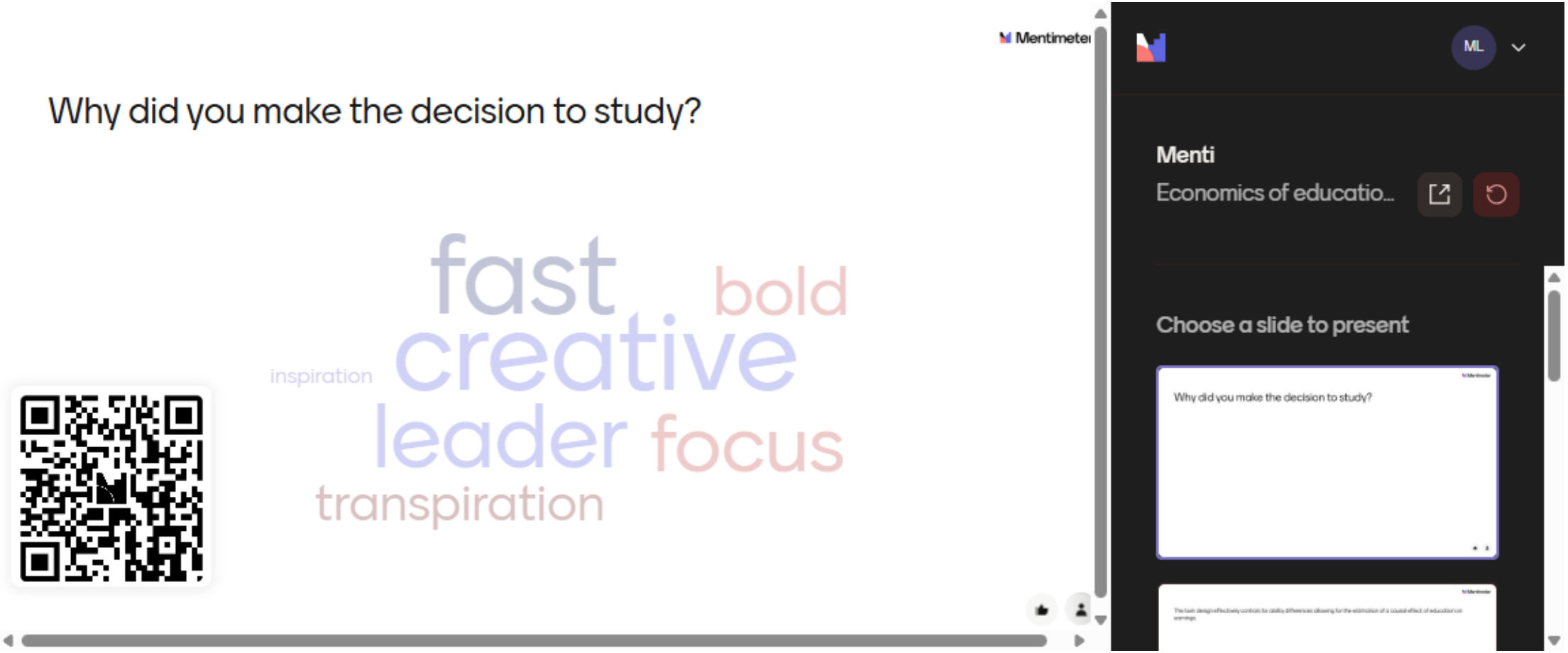
Outline for today

- Why do individuals want to receive education?

Main Readings

- Psacharopoulos & Patrinos (2018) “Returns to investment in education: a decennial review of the global literature.”
- Lovenheim & Turner (2018). Chapter 6 – The Returns to Education Investment
- Hanushek et al., (2011) Chapter 2 – Nonproduction Benefits of Education: Crime, Health, and Good Citizenship

Why do individuals want to receive education?



Why do individuals want to receive education?

- Some people don’t want to receive education
- They go because it’s mandatory

Why do individuals want to receive education?

Economic answer

- Financial benefits

Why do individuals want to receive education?

Modern economics answer

- To acquire skills (cognitive/non-cognitive/technical)
- Better health
- Life satisfaction
- Job prestige

….

- A better life

Roadmap

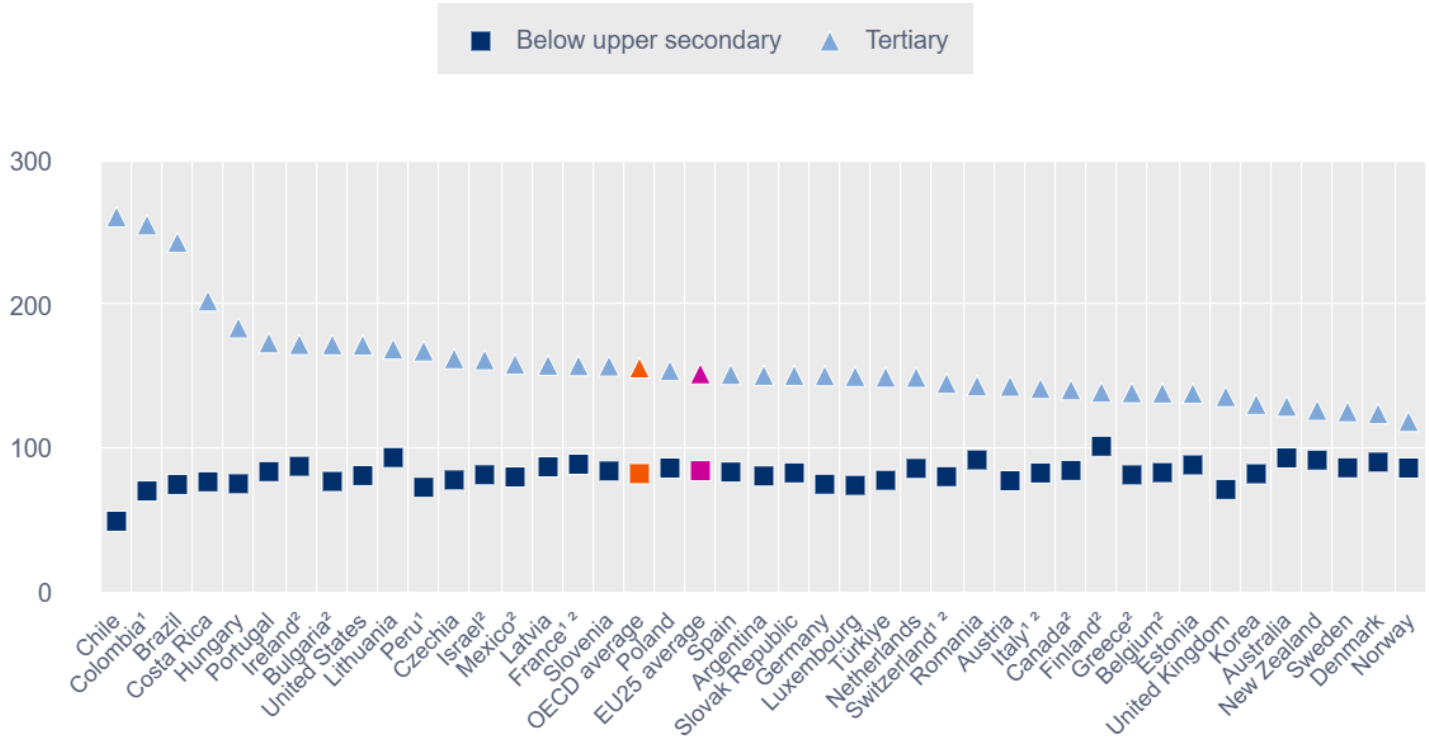
1. Financial benefits
    - Correlation
    - Mincer equation
    - Twin studies
    - Quasi-experimental studies
        - Quarter of birth
        - Proximity to schools
        - Compulsory schooling laws
2. Health benefits
3. Other non-monetary benefits

What does the evidence tell us?

Financial benefits

Relative earnings of workers compared to those with upper secondary attainment, by educational attainment (2022)

25-64 year-olds with income from employment (full-time full-year workers); upper secondary education = 100



Source: OECD. “Education at a Glance 2024.” *OECD*, 2024, www.oecd.org/en/publications/education-at-a-glance-2024_c00cad36-en.html.

What does the evidence tell us?

Financial benefits

- There is a large association between educational attainment and earnings
- Is this <u>causal</u>?

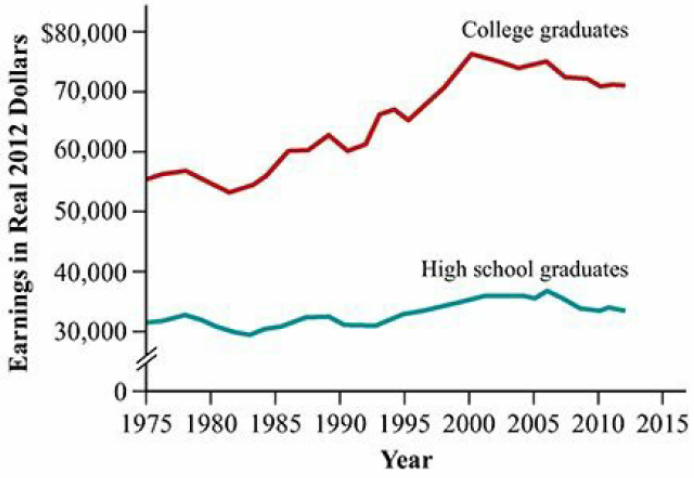


Figure 6.3
Lovenheim/Turner, *Economics of Education*, 1e
Data from: 1975–2012 Current Population Survey.

Figure 6.3 Annual Earnings Among Workers with a College Degree Versus a High School Diploma, Age 25 and Over

Earnings among high school graduates have remained flat since the mid-1970s, while earnings among those with a college degree have risen from about $55,000 per year to over $70,000 per year. The result is a substantial increase in the college earnings premium relative to high school graduates.

What does the evidence tell us?

Financial benefits

Mincer equation

- Do you spot any issues with this estimation technique?

$$Ln(Y) = \beta_0 + \beta_1 S + \beta_2 Exp + \beta_3 Exp^2 + U$$

In this equation, Y is earnings, S is years of education, Exp is the number of years since completion of formal schooling, and finally, U is a statistical error term. The Exp term often is referred to as *potential experience* because it measures the number of postschooling years a person could be working. This is a log-linear model that can be simply estimated using linear regression techniques.

What does the evidence tell us?

Financial benefits

Mincer equation

- Assumes linear relationship between education and earnings
- Hardly causal
- But easy to estimate!

$$Ln(Y) = \beta_0 + \beta_1 S + \beta_2 Exp + \beta_3 Exp^2 + U$$

In this equation, Y is earnings, S is years of education, Exp is the number of years since completion of formal schooling, and finally, U is a statistical error term. The Exp term often is referred to as *potential experience* because it measures the number of postschooling years a person could be working. This is a log-linear model that can be simply estimated using linear regression techniques.

What does the evidence tell us?

Financial benefits

Mincer equation

Based on 705 estimates, over the years 1950 to 2014, the private rate of return to an additional year of schooling is **8.8%**

Statistically significant decline of the returns of about **0.2 percentage points** for every additional year of schooling

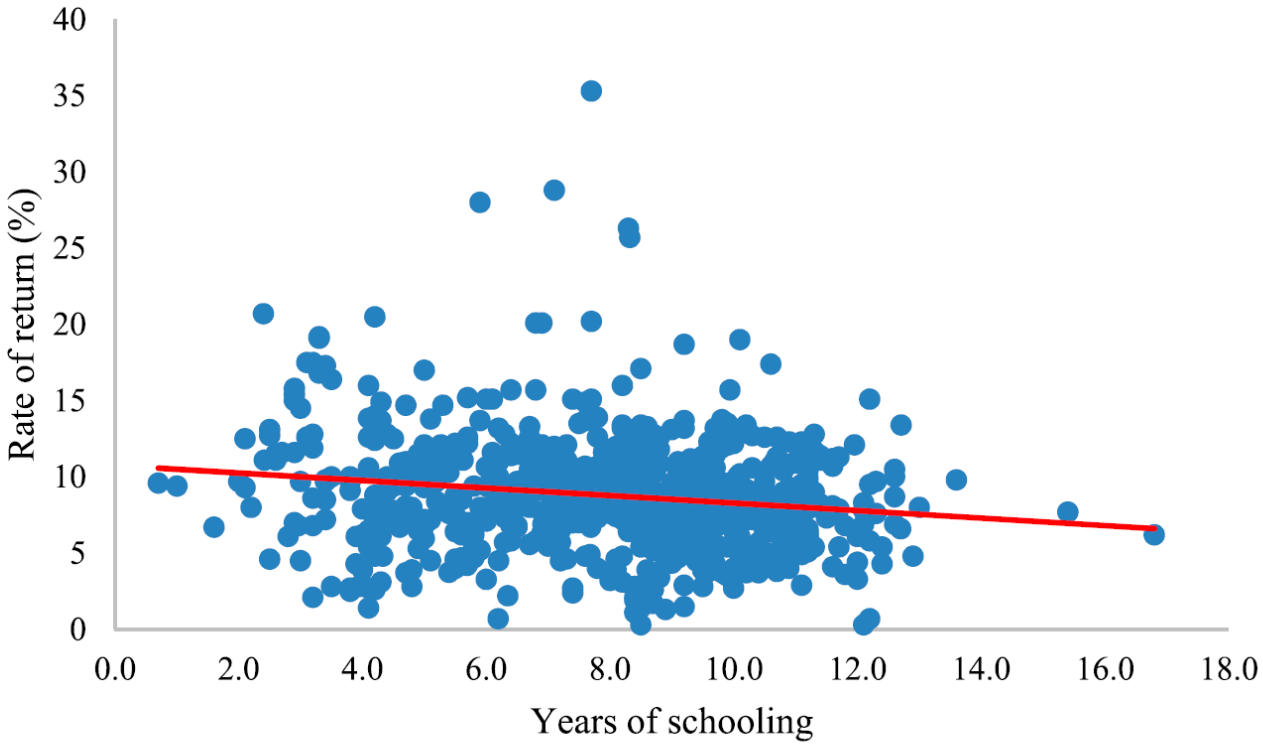


Figure 2. Return to schooling by years of schooling.

Source: Psacharopoulos & Patrinos (2018) "Returns to investment in education: a decennial review of the global literature."

What does the evidence tell us?

Financial benefits

Mincer equation

Stable returns over time; decrease is statistically insignificant

Considerable heterogeneity between studies

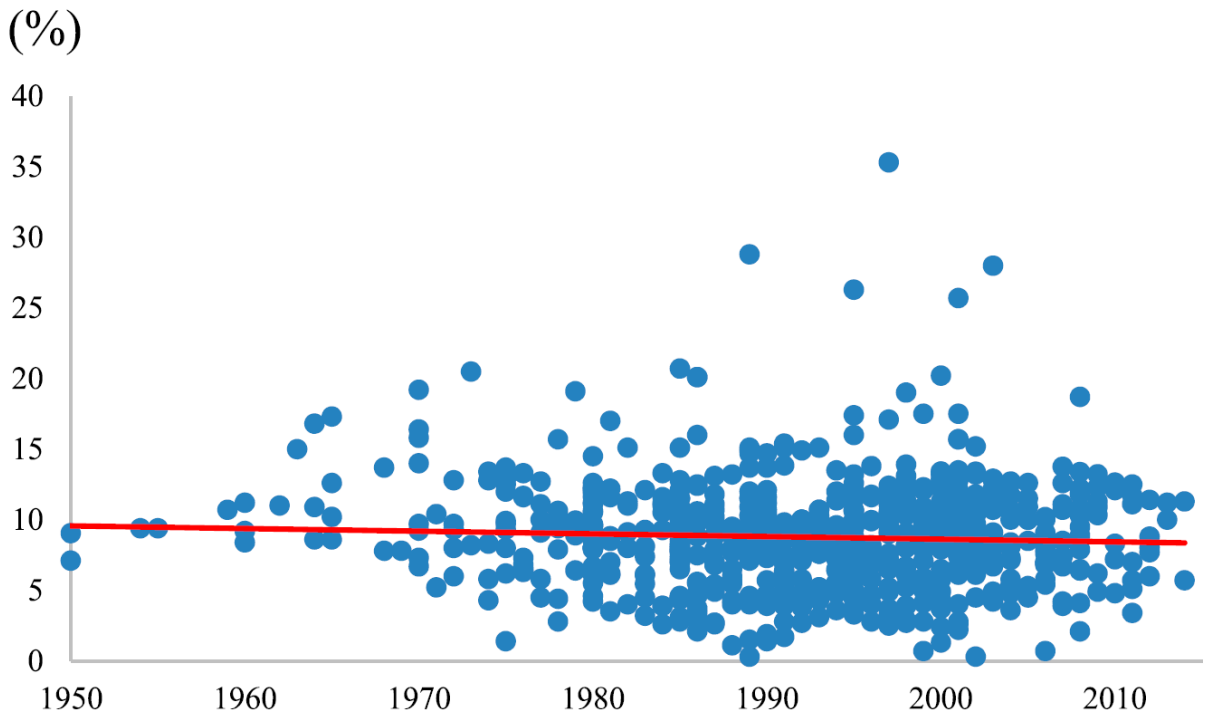


Figure 1. Rate of return to schooling over time.

Source: Psacharopoulos & Patrinos (2018) "Returns to investment in education: a decennial review of the global literature."

What does the evidence tell us?

Financial benefits

Twin studies

- Do you spot any issues with this estimation technique?

Typically, these studies examine monozygotic twins who are identical in terms of both family background and genetics. The idea is to relate the differences in earnings across twins to the differences in the amount of education they have. The underlying assumption that supports this strategy is that educational differences across twins are unrelated to any of their characteristics that also correlate with labor market earnings.

Source: Lovenheim, M., & Turner, S. E. (2018). Economics of education. Worth Publishers/Macmillan Learning. Chapter 6

What does the evidence tell us?

Financial benefits

Twin studies

- Does controlling for genes makes the estimate causal?
- The driver of educational differences is unknown

Typically, these studies examine monozygotic twins who are identical in terms of both family background and genetics. The idea is to relate the differences in earnings across twins to the differences in the amount of education they have. The underlying assumption that supports this strategy is that educational differences across twins are unrelated to any of their characteristics that also correlate with labor market earnings.

Source: Lovenheim, M., & Turner, S. E. (2018). Economics of education. Worth Publishers/Macmillan Learning. Chapter 6

What does the evidence tell us?

Financial benefits

Twin studies

- An additional schooling year increases earnings by 9-16%

Table 3—Ordinary Least-Squares (OLS), Generalized Least-Squares (GLS), Instrumental-Variables (IV), and Fixed-Effects Estimates of Log Wage Equations for Identical Twins[a]

Variable	OLS (i)	GLS (ii)	GLS (iii)	IV[a] (iv)	First difference (v)	First difference by IV (vi)
Own education	0.084 (0.014)	0.087 (0.015)	0.088 (0.015)	0.116 (0.030)	0.092 (0.024)	0.167 (0.043)
Sibling's education	—	—	−0.007 (0.015)	−0.037 (0.029)	—	—
Age	0.088 (0.019)	0.090 (0.023)	0.090 (0.023)	0.088 (0.019)	—	—
Age squared (÷100)	−0.087 (0.023)	−0.089 (0.028)	−0.090 (0.029)	−0.087 (0.024)	—	—
Male	0.204 (0.063)	0.204 (0.077)	0.206 (0.077)	0.206 (0.064)	—	—
White	−0.410 (0.127)	−0.417 (0.143)	−0.424 (0.144)	−0.428 (0.128)	—	—
Sample size:	298	298	298	298	149	149
R^2:	0.260	0.219	0.219	—	0.092	—

Notes: Each equation also includes an intercept term. Numbers in parentheses are estimated standard errors.

[a]Own education and sibling's education are instrumented for using each sibling's report of the other sibling's education as instruments.

Source: Ashenfelter, Orley, and Alan B. Krueger. "Estimates of the Economic Return to Schooling from a New Sample of Twins." *American Economic Review* 84, no. 5 (1994): 1157–1173.

What does the evidence tell us?

Financial benefits

Twin studies

- Replicates findings in larger sample
- Returns to schooling vary from 7.5-9,5%

Source: Rouse, C. E. (1999). Further estimates of the economic return to schooling from a new sample of twins. *Economics of Education Review*, *18*(2), 149-157.

Table 2
Ordinary least-squares (OLS), generalized least-squares (GLS), first-differenced, and instrumental variables (IV). Estimates of log wage equations for identical twins: Ashenfelter & Krueger's results and those obtained with the larger sample

	1991[a]				1991–1993, 1995[b]			
	OLS (1)	GLS (2)	First-differenced (3)	First-differenced by IV (4)	OLS (5)	GLS (6)	First-differenced (7)	First-differenced by IV (8)
Own education	0.084	0.087	0.092	0.167	0.105	0.101	0.075	0.095
	(0.014)	(0.015)	(0.024)	(0.043)	(0.008)	(0.009)	(0.017)	(0.027)
Age	0.088	0.090			0.097	0.098		
	(0.019)	(0.023)			(0.009)	(0.011)		
Age^2 (÷ 100)	−0.087	−0.090			−0.099	−0.099		
	(0.023)	(0.023)			(0.011)	(0.013)		
Female	−0.204	−0.206			−0.325	−0.326		
	(0.063)	(0.064)			(0.034)	(0.041)		
White	−0.410	−0.428			−0.082	−0.071		
	(0.127)	(0.128)			(0.060)	(0.072)		
Sample size	298	298	149	149	906	906	453	453
R^2	0.260	0.219	0.092		0.336	0.268	0.043	

Standard errors are in parentheses. All regressions include a constant. These regressions assume independent measurement errors.
[a] Source: Ashenfelter & Krueger (1994).
[b] Source: Twinsburg Twins Survey, 1991–93, 1995.

What does the evidence tell us?

Financial benefits

Twin studies

- Additional studies place estimates between 4-13%

Table 1
Summary of recent ln earnings relations for identical twins with control for measurement error in schooling

Source	Sample size, year, gender and race composition, and mean age	Preferred estimates[a]		Other variables in IV within MZ estimates
		IV Level	IV Within MZ	
1. Twinsburg, Ohio annual twins festival sample				
Ashenfelter and Krueger (1994), Tables 3 and 9	147 MZ twins pairs at 1991 Festival—54% female, 94% white, mean age 36.5	0.116	0.129	
Rouse (1998), Table 4	453 MZ twins pairs at 1991-3, 1995 Festivals—59% female, 92% white, mean age 37.9	0.111[b]	0.119	tenure, union, married
Ashenfelter and Rouse (1998), Table 3	333 MZ twins pairs at 1991-3 Festivals—59% female, 92% white, mean age 38.1	0.113[b]	0.100	tenure, union, married
2. United States' National Academy of Science–National Research Council twins sample of white veteran males born in 1917–27				
Behrman, Rosenzweig and Taubman (1994) Table 4	141 white male MZ pairs, 1973, mean age 51	0.101	0.050	
3. Australian Twin Register 1980–2 and 1988–9				
Miller, Mulvey and Martin (1995), Table 5	602 MZ twins pairs—51.4% female, mean age 36.8[c]	0.073	0.045	married

Source: Behrman, J. R., & Rosenzweig, M. R. (1999). "Ability" biases in schooling returns and twins: a test and new estimates. Economics of Education Review, 18(2), 159-167.

What does the evidence tell us?

Financial benefits

Twin studies

- Later study in the UK finds ~4-8% returns

Table 2—OLS, IV, and Within-Twin-Pair Estimates of the Return to Education for Identical U.K. Twins

	LFS	Twins							
		Without other covariates				Controlling for other covariates			
	Pooled	Pooled		Within pair		Pooled		Within pair	
	OLS (1)	OLS (2)	IV (3)	OLS (4)	IV (5)	OLS (6)	IV (7)	OLS (8)	IV (9)
Education	0.078	0.077	0.085	0.039	0.077	0.073	0.077	0.039	0.082
	(0.002)**	(0.011)**	(0.012)**	(0.023)	(0.033)*	(0.011)**	(0.018)**	(0.024)	(0.036)*
Age	0.058	0.078	0.077			0.059	0.059		
	(0.004)**	(0.021)**	(0.021)**			(0.024)*	(0.024)*		
Age^2 (÷100)	−0.062	−0.097	−0.095			−0.001	−0.001		
	(0.049)**	(0.027)**	(0.027)**			(0.000)**	(0.000)**		
London and South East						0.065	0.066	0.110	0.124
						(0.054)	(0.055)	(0.122)	(0.123)
Married						−0.004	−0.004	−0.053	−0.048
						(0.059)	(0.059)	(0.091)	(0.092)
Tenure (years)						0.012	0.012	−0.001	0.001
						(0.003)**	(0.003)**	(0.006)	(0.006)
Part time						−0.091	−0.089	−0.099	−0.102
						(0.065)	(0.065)	(0.097)	(0.098)
Observations	4,398	428	428	214	214	374	374	187	187
R^2	0.31	0.15	0.15	0.01	0.0009	0.21	0.21	0.03	0.009

Notes: Pooled twins regressions are based on $\log w_{if} = \alpha + \beta S_{if} + \gamma X_{if} + \varepsilon_{if}$ where w_{if} is the wage of twin i in family f, S_{if} is years of schooling, and X_{if} other covariates where indicated. Within-pair regressions are based on $\log w_{if} - \log w_{2f} = \beta(S_{1f} - S_{2f}) + \gamma(X_{1f} - X_{2f})$. Pooled estimates in column (1) are for all working women reporting a wage in the 1999 U.K. Labour Force Survey. Standard errors are in parentheses. Columns (1), (2), (3) and (6), (7) include a constant (not reported), the other columns exclude a constant. For the pooled IV estimates, twin 1's education is instrumented by twin 2's report of twin 1's education and vice versa. For the within-twins IV estimates, the difference in self-reported education is instrumented by the difference in the co-twin's report of the other's education.

* Significant at the 5-percent level.

** Significant at the 1-percent level.

Source: Bonjour, D., Cherkas, L. F., Haskel, J. E., Hawkes, D. D., & Spector, T. D. (2003). Returns to education: Evidence from UK twins. *American Economic Review*, *93*(5), 1799-1812.

What does the evidence tell us? Twin studies

Mentimeter

The twin design effectively controls for ability differences allowing for the estimation of a causal effect of education on earnings.

0	0
True	False

What does the evidence tell us?

Financial benefits

Twin studies

- Studies estimate a return on education of about 4-16%
- Controlling for genetic differences still yields a positive education-earnings association
- We can’t be sure about the causal effect of education yet

What does the evidence tell us?

Financial benefits

Quasi-Experimental Studies

- Quarter of birth
- Proximity to college
- Compulsory schooling laws

What does the evidence tell us?

Financial benefits

Quasi-Experimental Studies

Quarter of birth

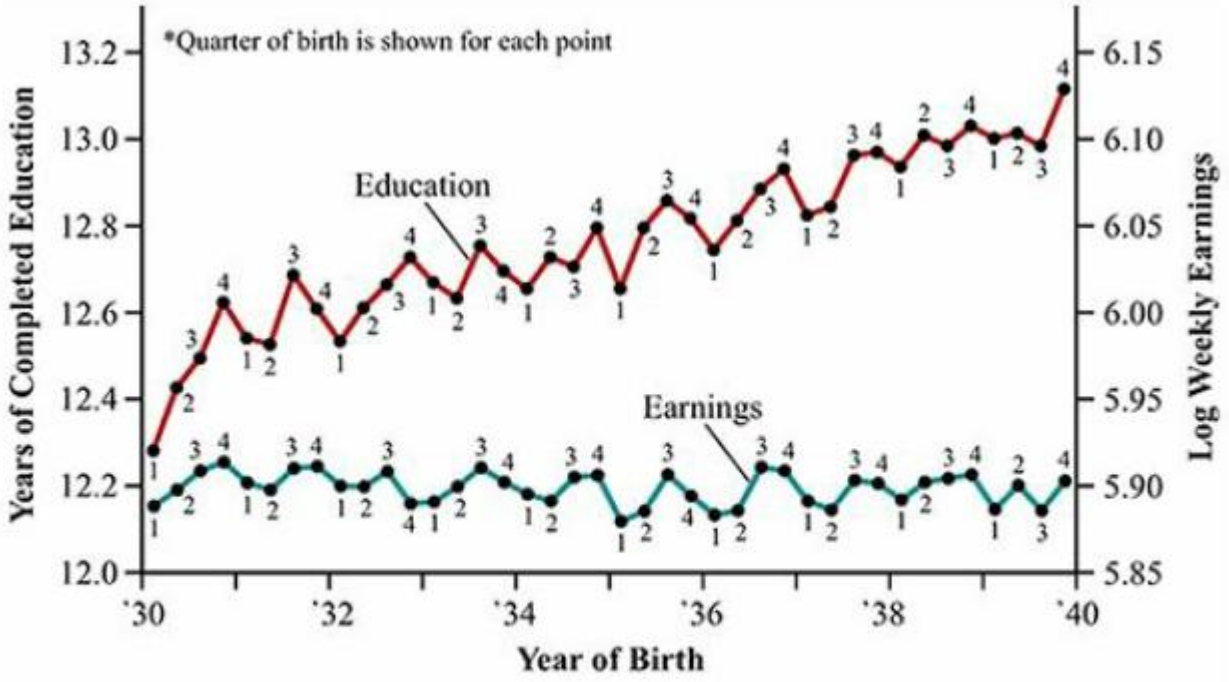


Figure 6.5
Lovenheim/Turner, *Economics of Education*, 1e
Data from: Figure I, Angrist and Krueger (1991) and Figure V, Angrist and Krueger(1991).

Source: Angrist, J. D., & Krueger, A. B. (1991). Does compulsory school attendance affect schooling and earnings?. *The quarterly journal of economics*, *106*(4), 979-1014.

What does the evidence tell us?

Financial benefits

Quasi-Experimental Studies

Quarter of birth

- Children who are born at the beginning of the year can drop out of school earlier and thus have less schooling on average
- Quarter of birth is thus an instrument for schooling and it correlates with having and correlates negatively with earnings
- Returns estimated to be ~10%

Source: Bound, J., & Jaeger, D. A. (2000). Do compulsory school attendance laws alone explain the association between quarter of birth and earnings?. In *Research in labor economics* (pp. 83-108).

What does the evidence tell us?

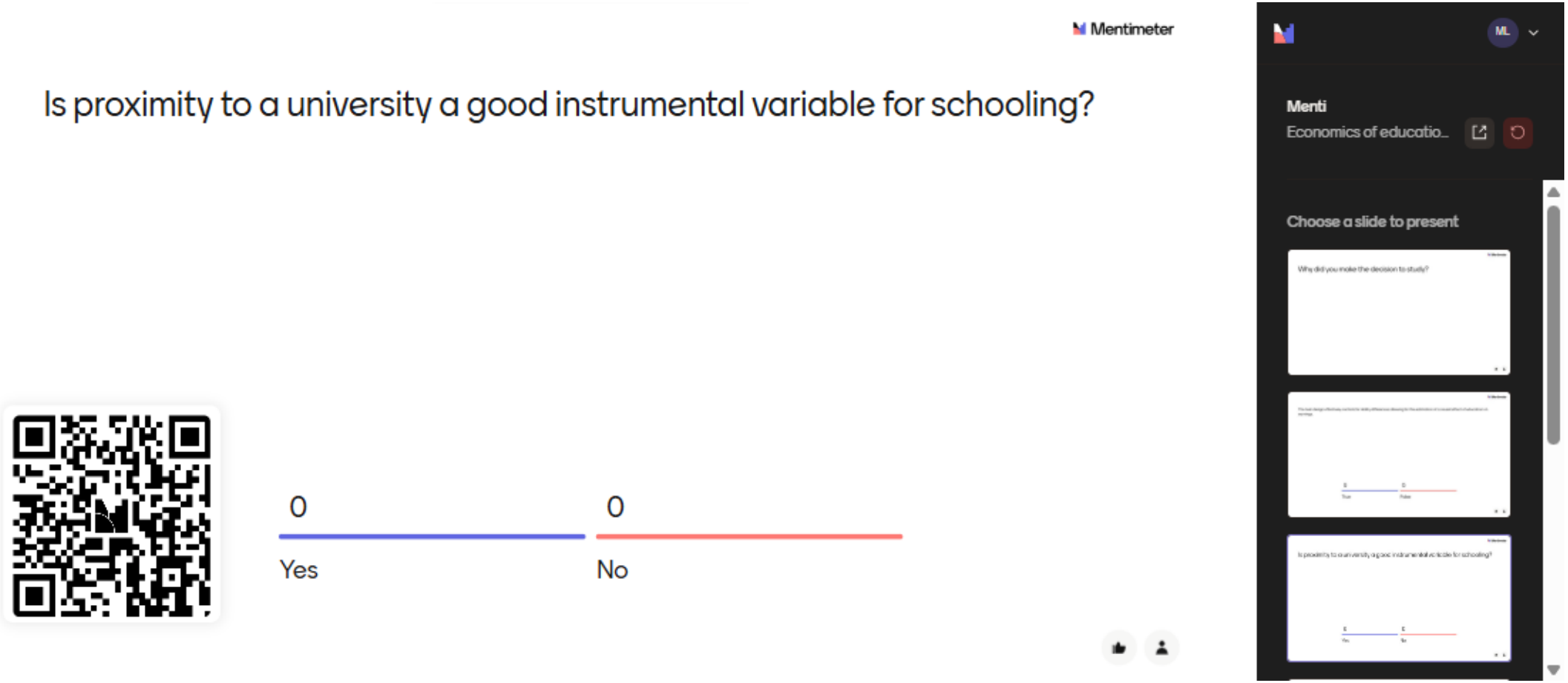
Financial benefits

Quasi-Experimental Studies

Living in proximity to college

- What do you think about this instrument?

What does the evidence tell us?



What does the evidence tell us?

Financial benefits

Quasi-Experimental Studies

Living in proximity to college

- What do you think about this instrument?
- Not a very credible instrument, as parents select the place of residency based on being urban vs rural, job opportunities, sector of employment, or household resources

What does the evidence tell us?

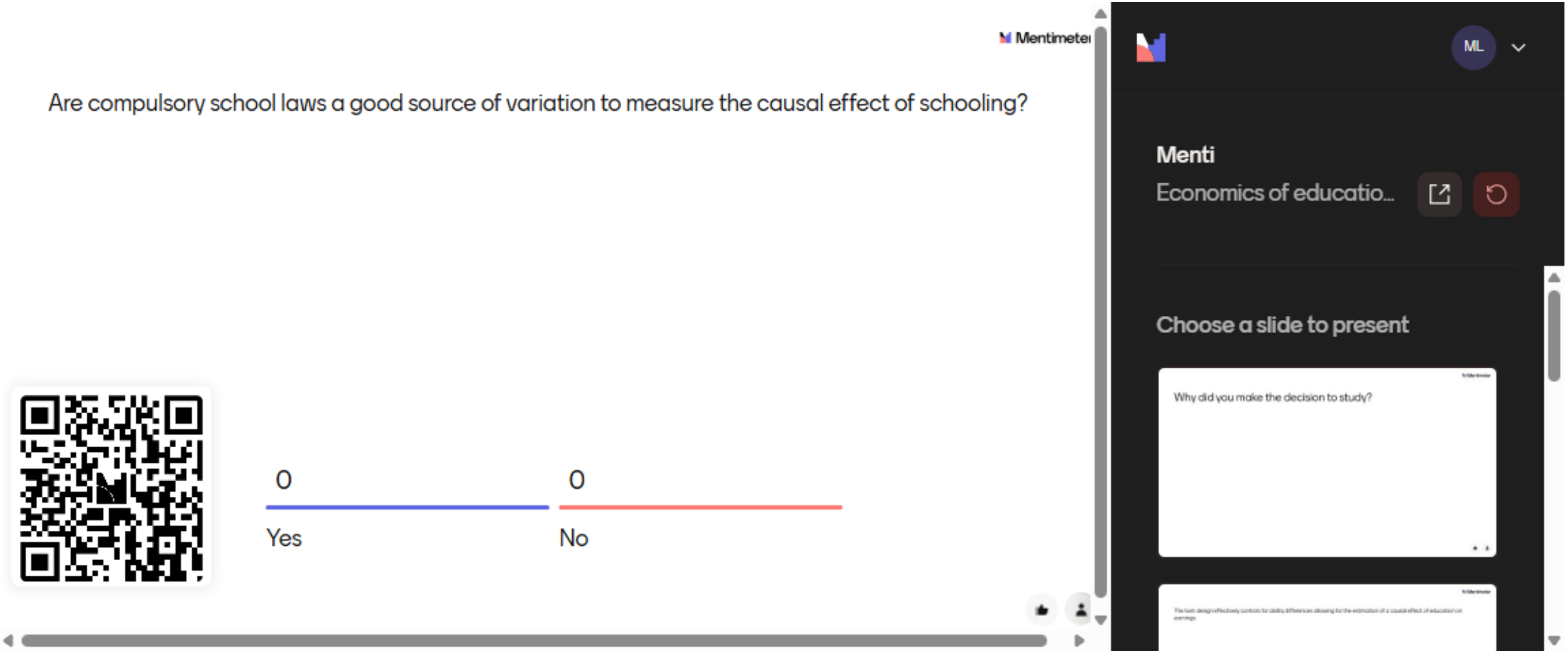
Financial benefits

Quasi-Experimental Studies

Compulsory schooling laws

- What do you think about method?

What does the evidence tell us?



What does the evidence tell us?

Financial benefits

Quasi-Experimental Studies

Compulsory schooling laws

- What do you think about this method?
- Definitely one of the most accepted methodologies
- Governmental based decision, likely unrelated with personal characteristics, no opportunity for selection into treatment

What does the evidence tell us?

Financial benefits

Quasi-Experimental Studies

Compulsory schooling laws

- UK sample
- 14 to 15 years

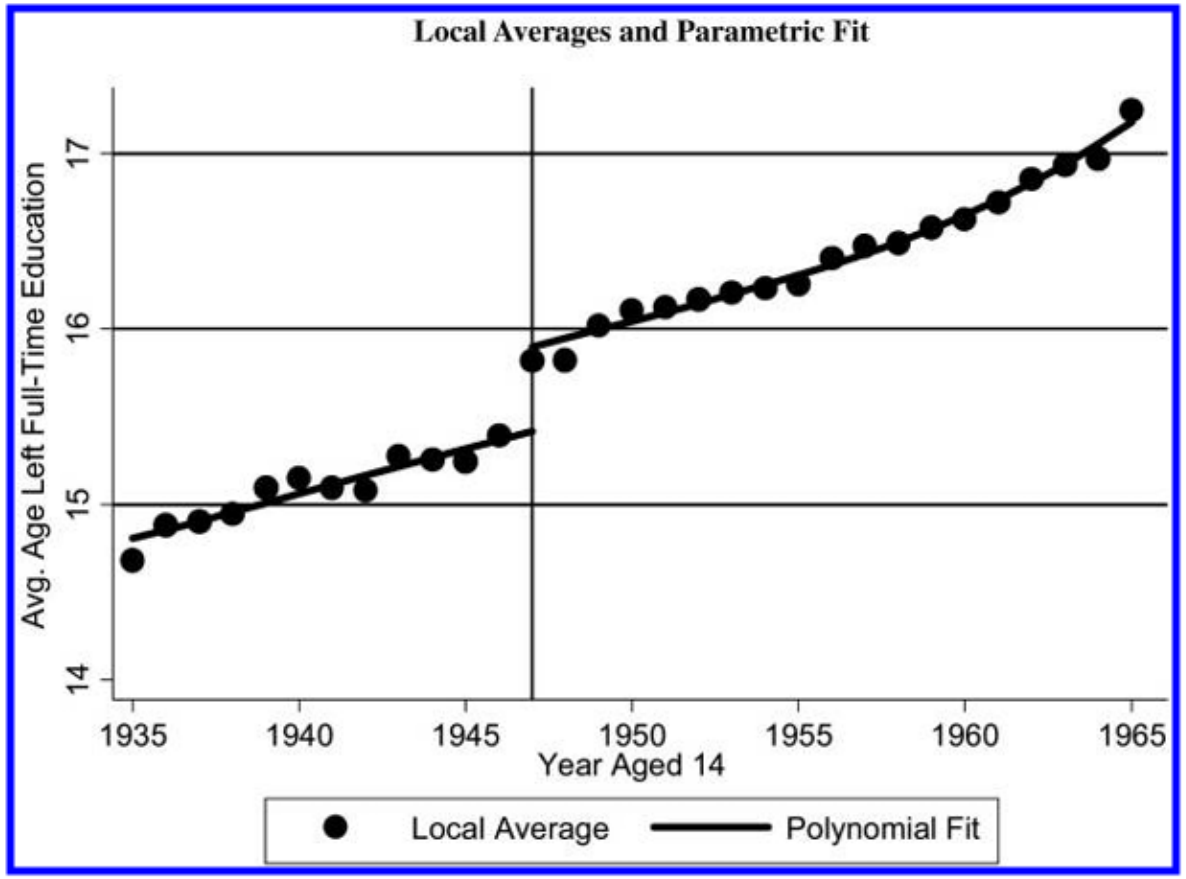


FIGURE 4. AVERAGE AGE LEFT FULL-TIME EDUCATION BY YEAR AGED 14 (*Great Britain*)

Note: Local averages are plotted for British-born adults aged 32 to 64 from the 1983 to 1998 General Household Surveys. The curved line shows the predicted fit from regressing average age left full-time education on a birth cohort quartic polynomial and an indicator for the school-leaving age faced at age 14. The school-leaving age increased from 14 to 15 in 1947, indicated by the vertical line.

Source: Oreopoulos, P. (2006). Estimating average and local average treatment effects of education when compulsory schooling laws really matter. American Economic Review, 96(1), 152-175.

What does the evidence tell us?

Financial benefits

Quasi-Experimental Studies

- **Compulsory schooling laws**
- One additional year of schooling leads to earnings 10-14% higher

Source: Oreopoulos, P. (2006). Estimating average and local average treatment effects of education when compulsory schooling laws really matter. American Economic Review, 96(1), 152-175.

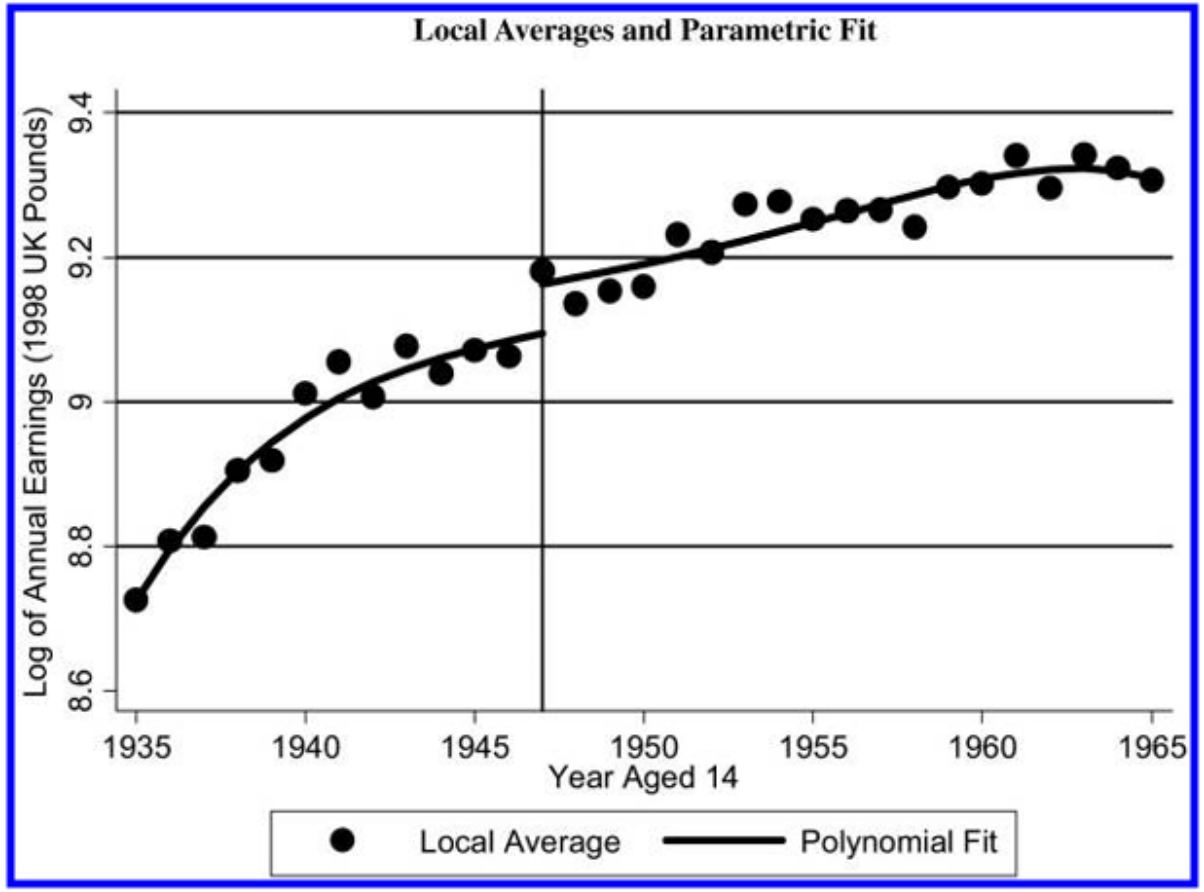


FIGURE 6. AVERAGE ANNUAL LOG EARNINGS BY YEAR AGED 14 (*Great Britain*)

Note: Local averages are plotted for British-born adults aged 32 to 64 from the 1983 to 1998 General Household Surveys. The curved line shows the predicted fit from regressing average log annual earnings on a birth cohort quartic polynomial and an indicator for the school-leaving age faced at age 14. The school leaving age increased from 14 to 15 in 1947, indicated by the vertical line. Earnings are measured in 1998 U.K. pounds using the U.K. retail price index.

What does the evidence tell us?

Mentimeter

What was the methodology used in this paper?

- Instrumental variable
- Difference-in-difference
- Regression discontinuity
- Regression with controls

What does the evidence tell us?

Why are compulsory schooling laws sometimes referred to as “instruments”

- Fuzzy instead of sharp regression discontinuity

What does the evidence tell us?

Financial benefits

Quasi-Experimental Studies

Compulsory schooling laws

- New study, focusing on same policy
- Re-analysed same data (General Household Survey, GHS)
- Re-analysed in new data, New Earnings Survey Panel Dataset (NESPD)
- New data has more detailed income information

Source: Devereux, P. J., & Hart, R. A. (2010). Forced to be rich? Returns to compulsory schooling in Britain. *The Economic Journal*, *120*(549), 1345-1364.

What does the evidence tell us?

Financial benefits

Quasi-Experimental Studies

Compulsory schooling laws

- New study, focusing on same policy
- Average return to schooling of about 3% using a similar specification and same data
- First author corrects his work

Source: Devereux, P. J., & Hart, R. A. (2010). Forced to be rich? Returns to compulsory schooling in Britain. *The Economic Journal*, *120*(549), 1345-1364.

What does the evidence tell us?

Financial benefits

Quasi-Experimental Studies

Compulsory schooling laws

- Using the new data set, findings are as follow

Source: Devereux, P. J., & Hart, R. A. (2010). Forced to be rich? Returns to compulsory schooling in Britain. *The Economic Journal*, *120*(549), 1345-1364.

What does the evidence tell us?

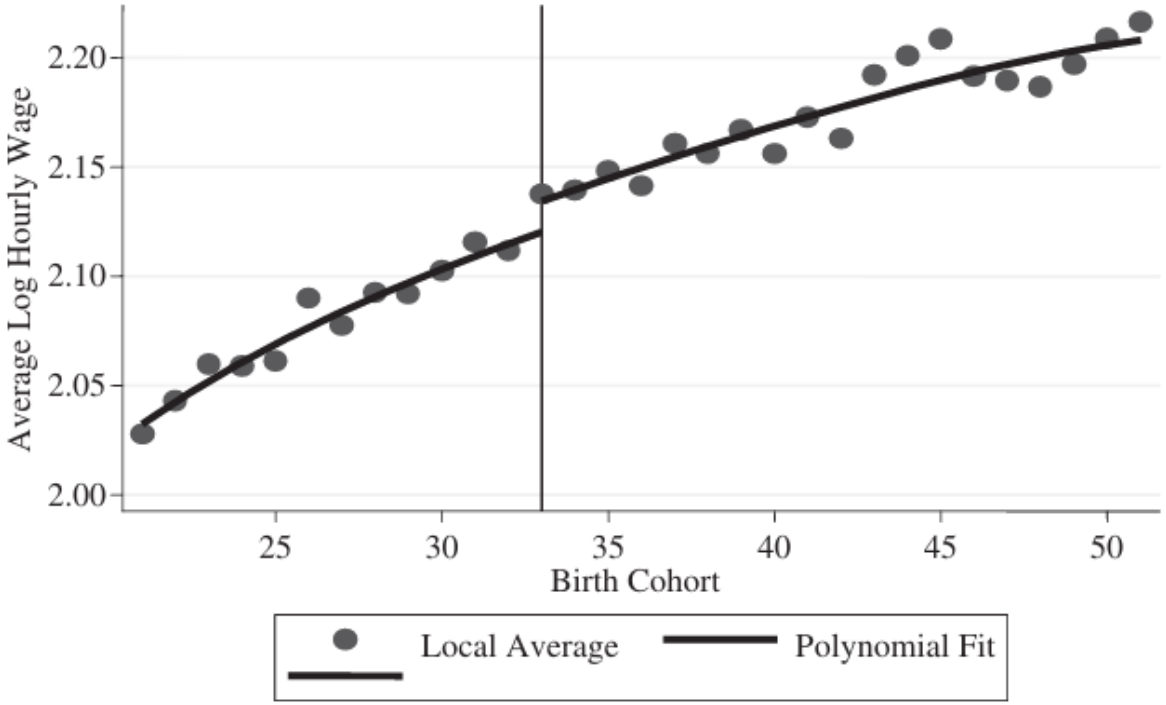


Fig. 8. *Average Hourly Wage by Year-of-Birth for Men* (NESPD)

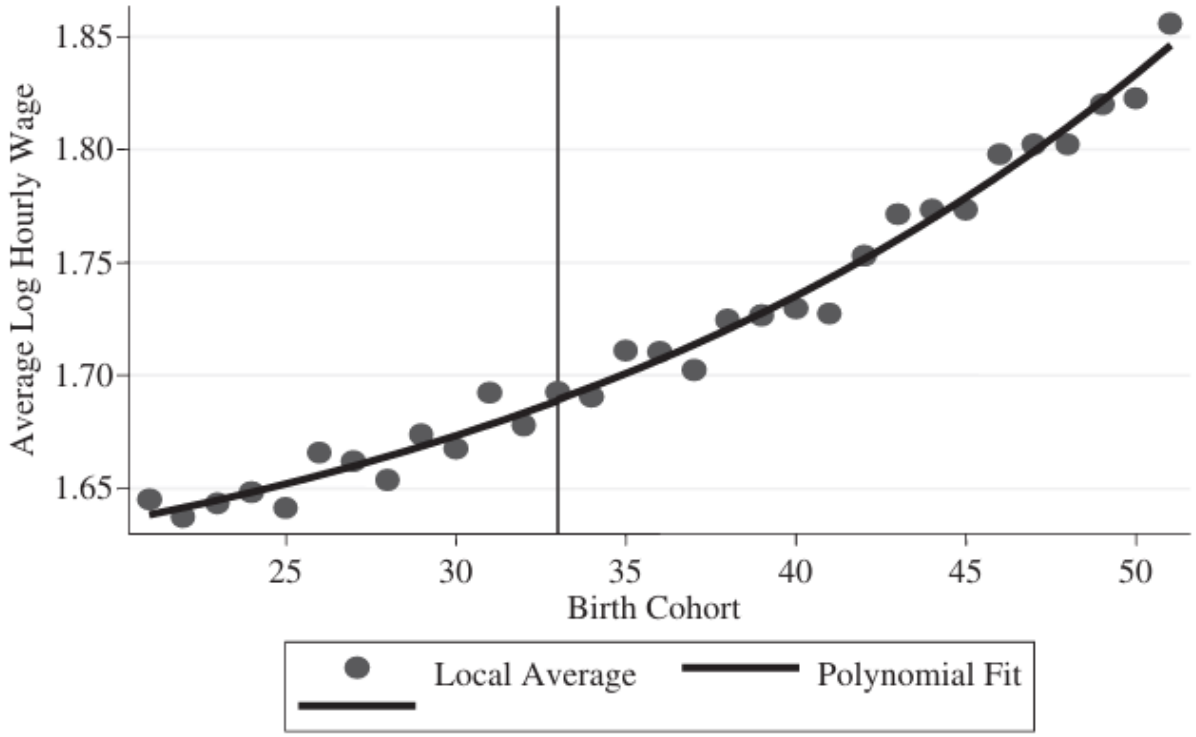


Fig. 7. *Average Hourly Wage by Year-of-Birth for Women* (NESPD)

Source: Devereux, P. J., & Hart, R. A. (2010). Forced to be rich? Returns to compulsory schooling in Britain. *The Economic Journal*, *120*(549), 1345-1364.

What does the evidence tell us?

Financial benefits

Quasi-Experimental Studies

- **Compulsory schooling laws**
- Much lower estimates using same and new data
- Zero returns for women and 3 - 4% for men

Source: Devereux, P. J., & Hart, R. A. (2010). Forced to be rich? Returns to compulsory schooling in Britain. *The Economic Journal*, *120*(549), 1345-1364.

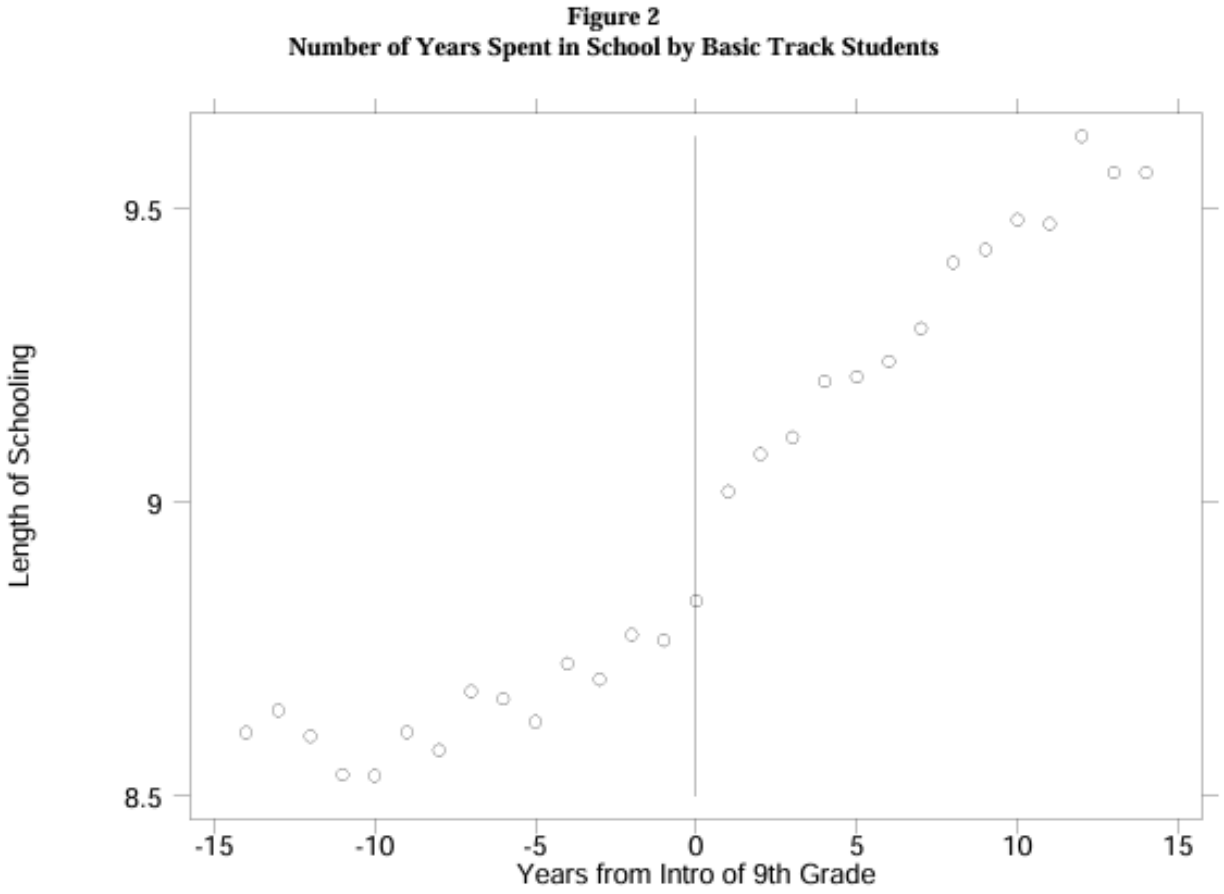
What does the evidence tell us?

Financial benefits

Quasi-Experimental Studies

Compulsory schooling laws

- Increase in compulsory schooling in West Germany
- 8th to 9th grade



Source: Pischke, J. S., & Von Wachter, T. (2008). Zero returns to compulsory schooling in Germany: Evidence and interpretation. *The Review of Economics and Statistics*, *90*(3), 592-598.

What does the evidence tell us?

Financial benefits

Quasi-Experimental Studies

Compulsory schooling laws

- Ols positive results
- IV Zero returns

Table 4
Log Wage Regressions
(Standard Errors in Parentheses)

Independent Variable	OLS (1)	OLS (2)	OLS (3)	IV (4)	IV (5)	IV (6)
Qualification and Career Survey						
Years of Education	0.072 (0.001)	0.064 (0.005)	0.061 (0.001)	0.007 (0.013)	0.019 (0.012)	0.032 (0.013)
Micro Census						
Years of Education	0.083 (0.0004)	0.079 (0.002)	0.074 (0.001)	0.005 (0.006)	0.005 (0.005)	0.010 (0.005)
Female Dummy	✓	✓	✓	✓	✓	✓
Quartic in Experience	✓	✓				
Quartic in Age			✓	✓	✓	✓
Year Dummies	✓	✓	✓	✓	✓	✓
State of Residence Dummies	✓	✓	✓	✓	✓	✓
State Specific Cohort Trends					✓	✓
State Specific Cohort Trends Squared						✓
Year of Birth Dummies		✓	✓	✓	✓	✓

Note: Number of observations is 54,126 in the QaC and 750,416 in the Micro Census. Years of education is instrumented with a dummy for cohorts with a 9th grade in basic track in the state in columns (4) to (6). Standard errors are adjusted for clusters at the state * year of birth level.

Source: Pischke, J. S., & Von Wachter, T. (2008). Zero returns to compulsory schooling in Germany: Evidence and interpretation. *The Review of Economics and Statistics*, *90*(3), 592-598.

What does the evidence tell us?

Financial benefits

Quasi-Experimental Studies

- **Compulsory schooling laws**

A meta- analysis that includes papers that have

a) a change in compulsory schooling laws, or in similar legal or institutional changes such as month of birth; and

b) adult outcomes that can be translated into earnings

Source: Clark, Gregory; Nielsen, Christian Alexander Abildgaard. The returns to education: a meta-study. EHES Working Paper, 2024.

What does the evidence tell us?

Financial benefits

Quasi-Experimental Studies

- **Compulsory schooling laws**

Average estimate is **8.5%** gain in earnings for each year of education

Correcting for publication bias, suggests an average return of **3%**

Source: Clark, Gregory; Nielsen, Christian Alexander Abildgaard. The returns to education: a meta-study. EHES Working Paper, 2024.

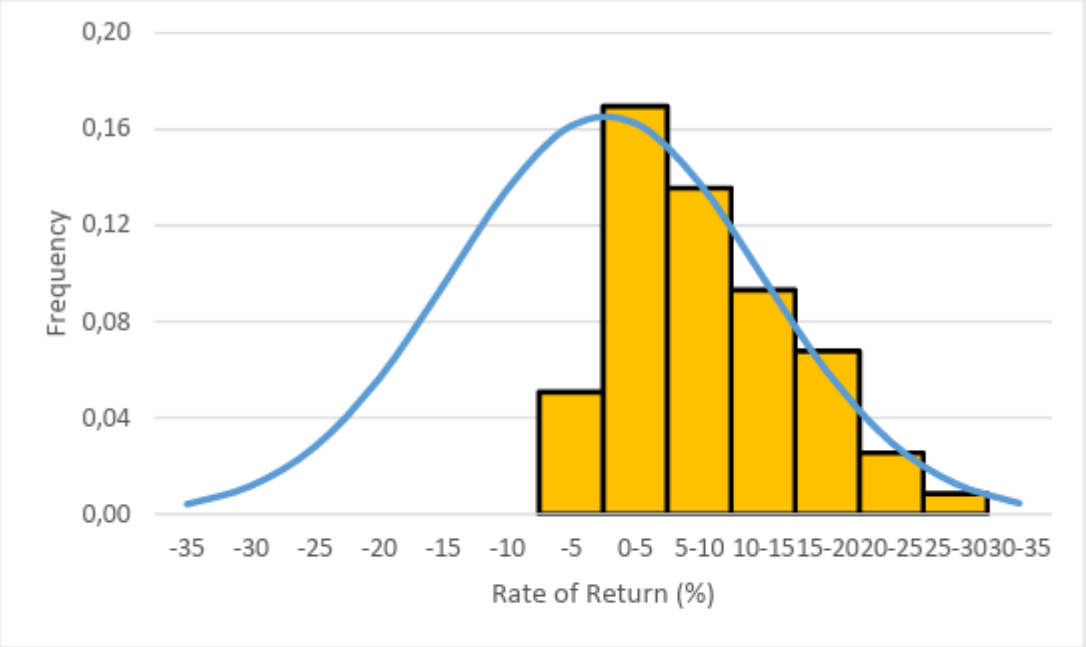


Figure 7: Relative Frequency of Estimated Rates of Return fitted to Normal Distribution

Notes: The figure shows 65 rates of return to an extra year of education reported in the meta-study versus the best fitting normal distribution of returns for reported returns greater than 0%. One outlier observation with a return greater than 50% was omitted.

Source: Meta-study dataset ("Clark_Nielsen_returns_to_education.dta" in online appendix).

What does the evidence tell us?

Financial benefits

- Lots of evidence suggesting a positive return on educational investment
- Some heterogeneity, with few studies pointing to null effects

Why is that?

What does the evidence tell us?

Financial benefits

- Education does not always translate to more skills
- Results might depend on level, quality of education, labour market, discrimination,…
- Although on average the effect is positive, it is unlikely that a single value exists
- Context matters

What does the evidence tell us?

Non-financial benefits: health

Consider the problem: The government says that they are looking for ways to reduce health care costs, and they have heard that investing on education can do so. They ask their advisors to look at the literature and answer the question: Does an increase in education improve health outcomes?

1. Get together in groups of 4
2. Open your book: Hanushek, E. A., Finis Welch, Machin, S., & Ludger Woessmann. (2011). *Handbook of the economics of education*. North-Holland. Chapter 2
3. Consider research summarized in pages 233-239 that investigates the effect of schooling on health outcomes
4. You have **15minutes** to write a reply to the government (1-2 sentences)

What does the evidence tell us?

Non-financial benefits: health

Table 2.4 Regression Discontinuity and Instrumental Variable Studies of the Effects of Years of Schooling on Mortality and Survival

Study	Data	Sample	Instrument	Important Features of Estimation Specification	Mortality or Survival Measure	Estimated Effect: No Instruments	Estimated Effect: RD/IV Methods
Mazumder (2008, 2010)	US censuses (1960–2000)	Native whites born 1901–1925	State-specific changes in compulsory schooling ages	2SLS with state-specific linear cohort trends	10-yr mortality rate	**–0.036** (0.003)	0.006 (0.031)
Lleras-Muney (2005, 2006)	US censuses (1960–1980)	Native whites born 1901–1925	State-specific changes in compulsory schooling ages	2SLS with region-specific linear cohort trends	10-yr mortality rate	**–0.036** (0.004)	**–0.063** (0.024)
Clark & Royer (2010)	Birth & mortality data, Office for National Statistics, England and Wales (1970–2007)	Residents of England & Wales in 1970–2004, born 1925–1938	National compulsory schooling age increase in 1947	RD with cohorts defined by quarter of birth*	Probability of dying between ages 45 and 69		Women 0.004; Men 0.012 (Reduced form not significant)
Albouy and Lequien (2009)	French Censuses and Enchantillon Demographique Permanent (various years 1968–1999), National Registers on Deaths (1968–2005)	French individuals born on first 4 days of October in 1920–1925 or 1950–1955 and alive in 1968	National compulsory schooling age increases for cohorts born in 1923 & 1953	RD with 1-yr cohort window around schooling age change	Survival from age 15 to 52		0.013 (0.011)
					Survival from age 45 to 82		0.063 (0.620)

Notes: Standard errors shown in parentheses. Estimates in bold are statistically significant at 0.05 level.
*Clark and Royer (2010) do not report IV estimates; values reported in table reflect RD estimated effect of 1947 reform on death rates between ages 45 and 69 divided by the effect of the reform on average years of completed schooling.

Source: Hanushek, E. A., Finis Welch, Machin, S., & Ludger Woessmann. (2011). *Handbook of the economics of education*. North-Holland. Chapter 2

What does the evidence tell us?

Non-financial benefits: health

Table 2.5 Recent US-Based Instrumental Variable Studies of the Effects of Schooling on Health

Study	Data	Sample	Instrument	Schooling Measure	Health Outcome	Estimated Effect: No Instruments		Estimated Effect: IV Methods	
Mazumder (2008)	Survey of Income and Program Participation (1984, 1986–1988, 1990–1993, 1996)	Native whites born 1901–1925	State-level compulsory schooling laws	Years of education	Self-reported health fair or poor	**–0.036** (0.001)		**–0.082** (0.034)	
					Number of nights in hospital last year	**–0.073** (0.0186)		–1.083 (0.767)	
					Trouble with stairs	**–0.025** (0.001)		–0.007 (0.032)	
					Health limitation	**–0.025** (0.0013)		**–0.074** (0.035)	
Oreopoulos (2006)	US censuses (1950–2000)	Natives born 1901–1956, ages 25–84 at time of survey	State-level compulsory schooling laws	Average years of education	Disability that limits personal care	**–0.014** (0.0003)		**–0.025** (0.006)	
					Disability that limits mobility	**–0.020** (0.0004)		**–0.043** (0.007)	
Kenkel, Lillard, & Mathios (2006)	National Longitudinal Survey of Youth, 1979 cohort	Born 1957–1964. Observed in 1998 (oversamples minorities & poor whites)	High school & GED require-ments & policies	High school graduate indicator		Women	Men	Women	Men
					Currently smoke	**–0.194** (0.030)	**–0.226** (0.029)	–0.102 (0.124)	**–0.229** (0.088)
					Obese	0.005 (0.029)	0.013 (0.026)	–0.021 (0.139)	–0.008 (0.082)

Source: Hanushek, E. A., Finis Welch, Machin, S., & Ludger Woessmann. (2011). *Handbook of the economics of education*. North-Holland. Chapter 2

What does the evidence tell us?

Non-financial benefits: health

Table 2.5 Recent US-Based Instrumental Variable Studies of the Effects of Schooling on Health—continued

Study	Data	Sample	Instrument	Schooling Measure	Health Outcome	Estimated Effect: No Instruments	Estimated Effect: IV Methods	
de Walque (2007)	National Health Interview Survey—Smoking Supplements (various years 1983–1995)	Natives born 1937–1956, aged 25+ at the time of the survey	Risk of induction in Vietnam War				Risk of induction	Risk of induction × Risk of being killed
				Number of years of education above high school	Currently smoke	**–0.040** (0.004)	–0.038 (0.020)	**–0.040** (0.019)
					Stopped smoking	**0.041** (0.002)	0.051 (0.031)	**0.066** (0.029)
				Indicator for college graduate or more	Currently smoke	**–0.173** (0.015)	–1.169 (0.642)	**–0.181** (0.087)
					Stopped smoking	**0.178** (0.010)	2.190 (1.395)	**0.297** (0.133)
Grimard and Parent (2007)	Current Population Survey Tobacco Supplements (1995, 1996, 1998, 1999)	White native-born citizens at least age 25, born 1935–1974	Vietnam draft cohort indicator (born 1945–1950)	Years of education	Currently smokes everyday	**–0.055** (0.002)	**–0.136** (0.024)	
					Ever smoked	**–0.043** (0.002)	**–0.111** (0.029)	

Notes: Estimates under "No Instruments" column are based on OLS estimation or Probit estimates (marginal effects reported). Estimates under "IV Methods" column report estimates from two-stage least squares or two-stage conditional maximum likelihood estimation (marginal effects reported). Standard errors are shown in parentheses. de Walque (2007) provides *t*-statistics rather than standard errors; standard errors are calculated as the reported estimates divided by reported *t*-statistics. Estimates in bold are statistically significant at 0.05 level.

Source: Hanushek, E. A., Finis Welch, Machin, S., & Ludger Woessmann. (2011). *Handbook of the economics of education*. North-Holland. Chapter 2

What does the evidence tell us?

Non-financial benefits: health

Table 2.6 Recent European Instrumental Variable Studies of the Effects of Years of Schooling on Health

Study	Data	Sample	Instrument	Health Outcome	Estimated Effect: No Instruments		Estimated Effect: IV/RD Methods	
Clark & Royer (2010)	Health Survey of England (1991–2004)	Residents of England in 1991–2004 (born 1918–1972)	National compulsory schooling age increases in 1947 and 1973	Self-reported health fair or Bad	**–0.081** (0.003)		–0.005 (0.014)	
				Long illness	**–0.033** (0.003)		0.023 (0.016)	
				Reduced activity	**–0.021** (0.002)		0.014 (0.012)	
				Obese	**–0.024** (0.003)		0.028 (0.016)	
				Hypertension	**–0.015** (0.003)		–0.025 (0.018)	
				Currently smoke	**–0.072** (0.003)		**–0.035** (0.014)	
				Physically active	**0.055** (0.004)		–0.001 (0.019)	
Powdthavee (2010)	Health Survey of England (1991–2007)	Residents of England in 1991–2007 (born 1929–1939 or 1952–1965)	National compulsory schooling age increases in 1947 and 1973		Women	Men	Women	Men
				Hypertension (1947 law change)	**–0.017**	**–0.015**	**–0.084**	**–0.070**
				Hypertension (1973 law change)	**–0.009**	**–0.008**	0.014	0.075
Oreopoulos (2006, 2008)	Annual General Household Survey for England (1983–1998) & Northern Ireland Continuous Household Surveys (1985–1998)	Native-born British & North Ireland residents, (born 1921–1951, ages 25–84 at time of survey)	Compulsory schooling age increases in United Kingdom (1947) & North Ireland (1957)	Self-reported poor health	**–0.037** (0.002)		0.007 (0.008)	
				Self-reported good health	**0.065** (0.002)		–0.010 (0.011)	

Source: Hanushek, E. A., Finis Welch, Machin, S., & Ludger Woessmann. (2011). *Handbook of the economics of education*. North-Holland. Chapter 2

What does the evidence tell us?

Non-financial benefits: health

Table 2.6 Recent European Instrumental Variable Studies of the Effects of Years of Schooling on Health—continued

Study	Data	Sample	Instrument	Health Outcome	Estimated Effect: No Instruments		Estimated Effect: IV/RD Methods	
							2SLS	RD
Silles (2009)	General Household Survey for England, Scotland, and Wales (1980–2004)	UK individuals (ages 25–60)	National compulsory schooling age increases in 1947 and 1973	Self-reported good health	**0.026** (0.001)		**0.045** (0.009)	**0.064** (0.003)
				No long-term illness	**0.008** (0.001)		**0.055** (0.009)	**0.075** (0.003)
				No activity-limiting illness	**0.008** (0.001)		**0.046** (0.008)	**0.051** (0.002)
					Women	Men	Women	Men
Kemptner, Jürges, & Reinhold (2010)	German Microcensus (1989, 1995, 1999, 2002, 2003)	Germans (born 1930–1960) living and educated in former West German states	Increased state-specific compulsory schooling from 8 to 9 (between 1949 and 1969)	Long-term illness	**–0.012** (0.002)	**–0.029** (0.001)	0.013 (0.024)	**–0.039** (0.019)
				Work-limiting illness	**–0.013** (0.001)	**–0.029** (0.001)	0.036 (0.022)	**–0.037** (0.017)
				Overweight	**–0.047** (0.002)	**–0.031** (0.001)	–0.016 (0.016)	**–0.034** (0.015)
				Obese	**–0.020** (0.001)	**–0.018** (0.001)	0.007 (0.012)	–0.028 (0.015)
				Ever smoked	**–0.005** (0.002)	**–0.026** (0.001)	0.014 (0.011)	–0.012 (0.013)
				Currently smoker	**–0.021** (0.002)	**–0.032** (0.001)	0.001 (0.01)	0.005 (0.012)
Brunello, Fabbri, and Fort (2009)	ECHP (1998), SHARE (2004), GSOEP (2002), BHPS (2003), Enquete sur la Sante (2003)	European women (from 10 countries) (ages 25–65), born within 7 yr of cohort first affected by school law changes	Increased compulsory schooling ages by country	Overweight	**–0.022** (0.001)		**–0.044** (0.021)	
				Obesity	**–0.012** (0.001)		**–0.012** (0.013)	

Notes: Estimates under "No Instruments" column are based on OLS estimation or Probit estimates (marginal effects reported). Estimates under "IV/RD Methods" column report estimates from two-stage least squares, IV probit estimation (marginal effects reported), or regression discontinuity (RD) methods. Standard errors are shown in parentheses. Estimates in bold are statistically significant at 0.05 level.

Source: Hanushek, E. A., Finis Welch, Machin, S., & Ludger Woessmann. (2011). *Handbook of the economics of education*. North-Holland. Chapter 2

What does the evidence tell us?

My response to the government based on those pages would be:

What does the evidence tell us?

Other non-financial benefits

- US data
- prestigious work
- less likely to have children when teen
- less likely to be divorced
- more likely to have voted
- More likely to be satisfied with life

Source: Oreopoulos, P., & Salvanes, K. G. (2011). Priceless: The nonpecuniary benefits of schooling. *Journal of Economic perspectives*, *25*(1), 159-184.

Table 2

Estimated Effects from an Extra Year of Compulsory Schooling

	[1]	[2]	[3]	[4]
Outcome	*Mean*	*Change in mean from 1 year of compulsory schooling before income controls*	*Change in mean from 1 year of compulsory schooling after income controls*	*Sample specification*
Log weekly income	3.06	0.131 (0.006)***	NA	Working
Log occupational prestige score	3.27	0.063 (0.003)***	0.046 (0.003)***	Working
Unemployed (*share*)	0.045	–0.005 (0.002)***	NA	Full
On welfare (*share*)	0.019	–0.015 (0.002)***	NA	Full
In jail (*share*)	0.027	–0.006 (0.003)*	NA	Black men 21–65 yrs old
In mental institute (share)	0.003	–0.001 (0.000)*	NA	Full
Probability of dying within 10 years[a]	0.11	–0.037 (0.006)***	NA	14 yrs old in 1914–1939 in 1960–1980 U.S. Censuses
First child born when teenager (*share*)	0.078	–0.008 (0.002)***	–0.008 (0.002)***	Women
Divorced (*share*)	0.137	–0.01 (0.002)***	–0.005 (0.003)***	Full
Oldest child behind grade level (*share*)	0.199	–0.032 (0.004)***	–0.026 (0.006)***	Parent matched to oldest 8–16 year old in household
Voted last national election (validated) (*share*)	0.49	0.09 (0.025)***	0.089 (0.027)***	Voting age 1978–2000 November Current Population Survey
Satisfied with life overall (*share*)	0.86	0.048 (0.010)***	0.035 (0.012)***	25–65 year olds born in UK from 1973–98 Eurobarometers

Source: Unless specified in column 4, all estimates are generated with a combined sample from the 1950–2000 U.S. Censuses and 2001–2007 American Community Surveys and include all 25–65 year-old native-born Americans aged 16 in 1915 or later with no more than a high school degree. The estimated effects from compulsory schooling on mortality are from Lleras-Muney (2004). The other estimates are by the authors.

What does the evidence tell us?

Non-financial benefits

- Studies that exploit quasi-natural experiments document positive returns of education on several health outcomes (self-reported health, disability, and smoking) but results are heterogeneous
- Evidence shows that education also positively affects other nonmonetary outcomes such as higher life satisfaction, later children, and more stable unions

What does the evidence tell us?

Overall

- The evidence shows that education improves one's life both financially and non-financially (health, life satisfaction)
- Individuals seem to have good incentives to pursue education
- Gains might not be for all; some studies find zero results + quasi-natural experiments tend to focus on basic education and on compliers
- It might not be necessarily irrational to not pursue higher levels of education for some people

Questions? Doubts?